Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.6 – Electrical Power

Unit 2: Electricity & Energy

The question: *a bulb and a kettle both run on electricity — so why does the kettle use so much more of it, so much faster?* The answer is **power**: how fast energy is transferred.

Learning intentions

All students will state the power formula and use it to calculate **power**.
Most students will use a second power formula, built from resistance.
Some students will calculate how much **energy** (and so how much it costs) to run something for a given time.

Part 1 — Power

Power is how much energy is transferred *every second*. It is measured in **watts (W)**, and it is linked to voltage and current by:

$$P = V \times I$$

1) **Calculate** the power of a bulb, if the voltage across it is **6 V** and the current through it is **0.5 A**.

Model answer: $P = V \times I = 6 \times 0.5 = 3\,W$.

Build it

Build a circuit with one bulb, an ammeter and a voltmeter. Record both readings and calculate the power.

$V =$ _____ V $I =$ _____ A $P = V \times I =$ _____ W

Check: if your bulb has a power rating printed on it, compare your answer to it — they should be close.

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Part 2 — A second way to find power

Last lesson's formula was $V = I \times R$. Substitute that straight into $P = V \times I$, in place of V:

$$P = V \times I = (I \times R) \times I = I^2 \times R$$

$$P = I^2 \times R$$

Same formula, same answer — just useful when you know the current and the resistance, but not the voltage.

2) **Calculate** the power of a **10 Ω** heating element, if a current of **2 A** flows through it.

Model answer: $P = I^2 \times R = 2^2 \times 10 = 4 \times 10 = 40\,W$.

Mixed practice

Calculate the power in each case. Show which formula you used.

Given	Power
$V = 12\,\text{V}$, $I = 3\,\text{A}$	$P = 36\,\text{W}$
$I = 1.5\,\text{A}$, $R = 8\,\Omega$	$P = 18\,\text{W}$
$V = 230\,\text{V}$, $I = 4\,\text{A}$	$P = 920\,\text{W}$

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Some students — energy and cost

Power is energy *per second*, so over a longer time:

$$\textbf{energy transferred} = \textbf{power} \times \textbf{time}$$

The more watts, and the longer it runs, the more energy it uses — and the more it costs.

3) **Calculate** how much more energy a **2000 W** kettle transfers in one minute than a **60 W** bulb left on for the same minute. Give both energies in joules (energy in joules = power in watts × time in seconds).

Model answer: *Kettle:* $2000 \times 60 = 120\,000\,J$. *Bulb:* $60 \times 60 = 3\,600\,J$. *The kettle transfers* $120\,000 - 3\,600 = 116\,400\,J$ *more — about 33 times as much, in the same minute.*

Part 3 — On your own

A string of **LED** fairy lights is rated at **24 W**. An older string of filament fairy lights, the same length, draws **1.2 A** from the **230 V** mains.

4) **Calculate** the power of the filament string, and state which string of lights is cheaper to leave on overnight, and by roughly how many times.

Model answer: *Filament string:* $P = V \times I = 230 \times 1.2 = 276\,W$. *The LED string (24 W) is far cheaper to run — about* $276 \div 24 \approx 11.5$ *times less power, so roughly a tenth of the cost for the same number of hours.*

HOMEWORK

Find the power rating (in W) on one appliance at home — it is often printed on a label or in the instructions. Write down the appliance and its power, then calculate how much energy (in joules) it would transfer if left running for **10 minutes** (600 seconds).